CSC 251 EXAM 2

STUDY GUIDE

EXAM STRUCTURE

- You will have the entire 75 minute meeting time to complete the exam.
- The exam will be **on paper only (no technology component)** and will include the following types of questions:
    - Short-answer
    - A short program, program fragments, or single lines of code to write or debug
    - Fill-in-the-blank with vocabulary terms
    - Multiple choice and T/F
    - Matching
- Materials
    - You may bring with you one 3-inch × 5-inch note card with any notes you would like to have for the exam written on **ONE SIDE** of the card. Note cards may not be shared between students, they must be handwritten, and they will collected when you turn in your exam.
    - If you have a note card, your name must be clearly printed on it.
    - If your note card is too large, it will be taken away or trimmed down. If you start taking the exam with a note card that's too large, your score on the exam will be penalized.
    - You may not use any notes during the exam.
    - You may not use any electronic devices such as cell phones or calculators during the exam.
    - You will be expected to write your answers in the space below each question. The instructor attempts to provide adequate space for each question. No additional pages or scratch paper should be added to your exam unless provided by the instructor.
    - The first page of the exam may be removed and used as a cover sheet while you complete your exam.
- If you must leave the classroom for a bathroom break, you must notify and leave your cell phone with the instructor. Only one student at a time may leave the room at a time. Late in the exam period, students may not be permitted to take a bathroom break.

EXAM TOPICS

Any topics covered during lectures 10-18 may be asked on this exam. Questions might require you write snippets of code, identify coding errors, predict the output of code, and explain concepts behind coding constructs and syntax.

- Data Science definition and data life cycle
- NumPy ndarrays (creation, arithmetic operations, indexing, slicing)
- pandas Series and DataFrames (creation, indexing, selecting, slicing/subsetting)
- Exploratory Data Analysis
- Types of plots used for data visualization of continuous and categorical variables, recognizing plot created through given code
- Causation vs. correlation, identifying and quantifying correlation
- Simple and multiple regression, underfitting vs. overfitting, predictive analysis
- Combining datasets through concatenation and merges
- Data cleaning techniques, issues with untidy data, date formats
- Machine learning concepts and types of models

EXAM PRACTICE

Be prepared to do the following:

- Given a chunk of code, diagnose errors in the code (not a full program)
- Understand the functionality of the packages described and used in class
- Create a NumPy array, Series, or DataFrame in various ways
- Write code to perform arithmetic and matrix operations with ndarrays to utilize vectorization
- Given a DataFrame, subset single or multiple rows/columns
- Given a dataset, determine the best type of plot to visualize specific characteristics of the data
- Recognize and demonstrate and understanding of code that creates various visualizations
- Describe steps in the Data Science Life Cycle
- Apply techniques to merge, clean, and wrangle data
- Interpret the results from the statsmodels linear and multiple regression analysis
    - Discuss the strength of the correlation between variables
    - Write a predictive equation
    - Demonstrate use of the predictive equation (no need for an exact answer without a calculator)
- Recognize and predict various ways to combine datasets
- Describe ways to handle missing values in a dataset
- Recognize or describe a machine learning model

SAMPLE PROBLEMS

The following are representative problems that you might encounter on the exam. Many of these will be familiar from the notes or lab.

1) Create a 1D NumPy ndarray containing the values 10, 20, 30, . . .5000.

2) Create a 100 x 200 ndarray containing all 0's.
3) Write a line of code to output the dimensions of your array from #7.
4) Apply ndarray operations to sample arrays.
5) Given the ndarray below, indicate the output:

```
two = np.array([[[1,2,3],[4,5,6]], [[7, 8, 9],[10,11,12]]])
two[1] = 0
print(two)
```

6) Given the ndarray below, indicate the output:

```
two = np.array([[10,20,30],[40,50,60], [70, 80, 90],[100,110,120]])
two_slice = two[2:, 1:2]
print(two_slice)
```

7) Given the following, what is the output if written in a Jupyter notebook?

```
data = pd.Series(['a', 'b', 'c'], index=[1, 3, 5])
data.loc[1]
data.iloc[1:3]
data[data > 'b']
```

8) Given a description and view of a DataFrame, be able to do the following:
    a. Write the dictionary used to create the DataFrame
    b. Add or delete a column
    c. Subset or filter the DataFrame
    d. Output the dataFrame shape and type

9) Indicate the best visualization for a univariate or bivariate description of a dataset. For example, for the tips dataset, which type of visualization best illustrates the number of parties dining on each day of the week?
10) Describe and interpret the visualization created by a line(s) of code. (You will NOT be expected to write code for visualizations.)
11) Qualitatively describe the strength of correlation from a visual representation.
12) Recognize and interpret the statistical output and visualization of single or multiple regression.
13) Write the resulting equation from single or multiple regression and demonstrate how to use it to calculate a predicted value. (Plug in values but no need to compute without a calculator.)
14) Describe and recognize DataFrame concatenation and various joins, provide code for each.
15) Know vocabulary!